
Chapter 5. Specifications

Important

Unless otherwise stated, all specifications apply after 30 minutes of instrument warm-up.

Important

Changes in the specification parameters are explicitly mentioned in the revision history of this document.

Important

Some specifications depend on the installed options. The options installed on a given instrument are listed in the Device tab of the LabOne user interface.

5.1. General Specifications

Table 5.1. General specifications

Parameter	Description
Storage temperature	+5°C to +65°C
Storage relative humidity	< 95%, non-condensing
Operating environment	IEC61010, indoor location, installation category II, pollution degree 2
Operating altitude	up to 2000 meters
Operating temperature	+5°C to +40°C
Operating relative humidity	< 90%, non-condensing
Specification temperature	+18°C to +28°C
Power consumption	<40 W
DC power inlet	12 V, 2 A Connector: Switchcraft 760BK, ID 2.5 mm, OD 5.5 mm
Power supply AC line	100–240 V ($\pm 10\%$), 50/60 Hz
Line power fuse	250 V, 2 A, fast, 5 x 20 mm, F 2 A L 250 V
Dimensions including bumper	28.3 x 23.2 x 10.2 cm 11.1 x 9.1 x 4.0 inch Rack mount on request
Weight including bumper	3.8 kg
Recommended calibration interval	2 years (see sticker on back panel)
Warranty	1 year, extensible

Table 5.2. Demodulators

Parameter	Description
Frequency range	DC to 500 kHz DC to 5 MHz, requires MF-F5M option
Number of demodulators	1 dual-phase (X, Y, R, Θ) 4 dual-phase, requires MF-MD option
Demodulator inputs	Signal Inputs (V/I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs
Filter time constant	336 ns to 83 s
Filter bandwidth (-3 dB)	276 μ Hz – 206 kHz (4th order filter)
Harmonics	1 – 1023
Filter slope	6, 12, 18, 24, 30, 36, 42, 48 dB/oct
Additional filtering	Sinc filter
Phase resolution	10 μ deg
Frequency resolution	1 μ Hz
Output sample rate on Auxiliary Outputs	612 kSa/s (for each auxiliary output), 18 bit, ± 10 V

Parameter	Description
Maximum transfer rate over 1 GbE	200 kSa/s (all demodulators), 48-bit full range
Maximum rate to store on local USB drive	50 kSa/s (all demodulators), 48-bit full range
Trigger modes for data transfer	Continuous, edge, gated

Table 5.3. Reference frequencies

Parameter	Description
External reference frequency range	1 Hz to 500 kHz 1 Hz to 5 MHz, requires MF-F5M option
External reference input	Auxiliary Inputs, Trigger Inputs, Auxiliary Outputs, Current Signal Input, Voltage Signal Input
Lock time for external reference	Typically less than max(100 cycles, 1.2 ms)
Number of external references	1; 2, requires MF-MD option
Internal reference frequency range	0 to 500 kHz 0 to 5 MHz, requires MF-F5M option

Table 5.4. Scope

Parameter	Description
Input channels	Signal Inputs (V,I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs, Trigger Outputs, Signal Output, Oscillator Phase. Demodulator R, Theta, X, Y, requires MF-DIG option
Scope modes	Time domain, frequency domain (FFT)
Number display channels	1; 2, requires MF-DIG option
Trigger channels	Signal Inputs (V,I), Auxiliary Inputs, Auxiliary Outputs, Trigger Inputs, Trigger Outputs. Demodulator R, Theta, X, Y, requires MF-DIG option
Trigger modes	Edge
Trigger hysteresis	Full input range
Pre-trigger	Full sample range
Sampling rates	1.8 kSa/s to 60 MSa/s
Vertical resolution	16 bit
Maximum number of samples per shot	16 kSa; 5 MSa, requires MF-DIG option
Minimum hold time	1 ms
Bandwidth limit mode, vertical resolution increase	Downsampling by averaging; increase vertical resolution up to 24 bit, requires MF-DIG option
Cursor math	Location, Area, Wave, Peak, Tracking, Histogram

Table 5.5. Spectrum

Parameter	Description
Center frequency range	0 to 500 kHz

Parameter	Description
	0 to 5 MHz, requires MF-F5M option
Spectrum modes	FFT(X+Y), FFT(R), FFT(Θ), FFT(f) and FFT((d Θ /dt)/2 π)
Statistical options	Amplitude, Spectral density, Power
Averaging modes	None, Exponential moving average
Maximum number of samples per spectrum	8 kSa
Maximum span	58 kHz
Window functions	Rectangular, Hann, Hamming, Blackman Harris
Cursor math	Location, Area, Tracking, Wave, Peak, Histogram

Table 5.6. Sweeper

Parameter	Description
Sweep parameters	Oscillator frequency, Demodulator phase, Auxiliary Offset, Signal Output Offset, etc.
Parameter sweep ranges	Full range, Linear and Logarithmic
Parameter sweep resolution	Arbitrary, defined by start/stop value and number of sweep points
Display parameters	Demodulator Output (X, Y, R, Θ , f), Auxiliary Input
Display options	Single Plot, Dual Plot (e.g. Bode Plot), Multi-trace
Statistical options	Amplitude, Spectral density, Power
Preset measurement modes	Parameter sweep, Noise amplitude measurement, Frequency response analyzer, 3-Omega-Sweep

Table 5.7. MF-IA Impedance Analyzer option

Parameter	Description
Frequency range	DC to 500 kHz DC to 5 MHz, requires MF-F5M option
Basic accuracy	0.05% (1 mHz to 500 kHz)
Basic temperature stability	<200 ppm/K
Test signal level	0 to 2.1 V _{rms} with monitoring
Bandwidth	276 μ Hz to 206 kHz
DC bias signal level	± 10 V (2 Terminal); ± 3 V (4 Terminal)
Compensation methods	SO, SOL, LLL, SL, L, OL
Impedance Z: range; basic accuracy	1 m Ω to 1 T Ω ; 0.05%
Admittance Y: range; basic accuracy	1 pS to 1 kS; 0.05%
Resistance Rs, Rp: range; basic accuracy	1 m Ω to 10 G Ω ; $\max(10 \mu\Omega, 0.05\%)^2$
Capacitance Cs, Cp: range; basic accuracy	10 fF to 1 F; $\max(10 \text{ fF}, 0.05\%)^2$
Inductance Ls, Lp: range; basic accuracy	100 nH to 1 H; $\max(10 \text{ nH}, 0.05\%)^2$

Parameter	Description
DC Resistance R_{DC} : range; basic accuracy	1 m Ω to 10 G Ω ; 2%
Reactance X: range; basic accuracy	1 m Ω to 10 G Ω ; 0.05%
Conductance G: range; basic accuracy	1 nS to 1 kS; max(100 nS, 0.05%)
Susceptance B: range; basic accuracy	1 nS to 1 kS; max(100 nS, 0.05%)
Loss coefficient D: range	10^{-4} to 10^4
Q factor: range	10^{-4} to 10^4

²Accuracy valid if parameter is the dominant value of the circuit representation.

Table 5.8. Voltage Signal Inputs

Parameter	Description
Connectors	2 BNC on front panel, single-ended or differential
Shield connectivity	Floating or ground
Maximum float voltage versus ground	± 1 V
Input impedance	50 Ω and 10 M Ω 27 pF for range ≥ 300 mV; 40 pF for range ≤ 100 mV
Input frequency range	DC to 500 kHz; DC to 5 MHz, requires MF-F5M option
Input A/D conversion	16 bit, 60 MSa/s
Input noise amplitude	2.5 nV/ $\sqrt{\text{Hz}}$ for frequencies > 1 kHz 7 nV/ $\sqrt{\text{Hz}}$ at 10 Hz 40 nV/ $\sqrt{\text{Hz}}$ at 1 Hz 3.3 mV input range; shorting cap on input
Input noise corner frequency	Typically 100 Hz for range ≤ 10 mV
Input bias current	Typically ± 10 pA, max ± 200 pA
Input full range sensitivity (10 V lock-in amplifier output)	1 nV to 3 V
Input AC ranges	± 1 mV, ± 3 mV, ± 10 mV, ± 30 mV, ± 100 mV, ± 300 mV, ± 1 V, ± 3 V
AC coupling cutoff frequency	1.6 Hz
Maximum DC offset for AC coupling	± 10 V
Input DC ranges	± 1 mV, ± 3 mV, ± 10 mV, ± 30 mV, ± 100 mV, ± 300 mV, ± 1 V, ± 3 V
Input gain inaccuracy	$< 1\%$ (< 2 MHz); for higher frequencies limited by analog input filter
Analog input filter (anti-aliasing)	1 dB suppression at 5 MHz, 3 dB at 12 MHz; 3rd order roll-off
Input amplitude stability	0.1%/°C
Input offset amplitude	$< \max(0.5 \text{ mV}, 1\% \text{ of range})$
Dynamic reserve	Up to 120 dB
Harmonic distortion	80 dBc for frequencies ≤ 100 kHz 65 dBc for frequencies ≤ 5 MHz

Parameter	Description
	carrier amplitude 1 dBFS
Coherent pickup	< -140 dB for frequencies ≤ 5 MHz and 50 Ω input impedance; < -180 dB for frequencies ≤ 100 kHz and 50 Ω input impedance

Table 5.9. Current Signal Input

Parameter	Description
Connector	BNC on front panel, float/ground
Shield connectivity	Floating or ground
Maximum float voltage versus ground	± 1 V
Input impedance	see Table 5.10
Input frequency range	0 to 500 kHz 0 to 5 MHz, requires MF-F5M option
Input A/D conversion	16 bit, 60 MSa/s
Input leakage current	± 10 pA
Input full range sensitivity (10 V lock-in amplifier output)	10 fA to 10 mA
Input gain inaccuracy	< 1% (for frequencies below 10% of the input bandwidth)
Input offset amplitude	1% of range
Input offset voltage	± 2.2 mV max; to shield of current input BNC connector
Dynamic reserve	up to 120 dB
Coherent pickup	< 90 G Ω for frequencies ≤ 5 MHz and 100 μ A input range < 140 T Ω for frequencies ≤ 100 kHz and 10 nA input range

Table 5.10. Current Signal Input: input ranges, transimpedance gain, bandwidth, input impedance, noise

Current input range	Transimpedance gain	Bandwidth (-3 dB)	Input impedance at DC	Input noise
10 mA	100 V/A	5 MHz	50 Ω	300 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
1 mA ¹	1 kV/A	5 MHz	50 Ω	200 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
100 μ A	10 kV/A	5 MHz	60 Ω	3.5 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
10 μ A ¹	100 kV/A	5 MHz	60 Ω	2.5 pA/ $\sqrt{\text{Hz}}$ at 100 kHz
1 μ A	1 MV/A	450 kHz	1 k Ω	200 fA/ $\sqrt{\text{Hz}}$ at 1 kHz
100 nA ¹	10 MV/A	450 kHz	1 k Ω	150 fA/ $\sqrt{\text{Hz}}$ at 1 kHz
10 nA	100 MV/A	2 kHz	160 k Ω	20 fA/ $\sqrt{\text{Hz}}$ at 100 Hz
1 nA ¹	1 GV/A	2 kHz	160 k Ω	15 fA/ $\sqrt{\text{Hz}}$ at 100 Hz

¹Range only available on MF Instruments with serial numbers MF-DEV3200 and higher.

Table 5.11. Signal Output

Parameter	Description
Connectors	2 BNC on front panel, single ended and differential
Output impedance	50 Ω
Output frequency range	DC to 500 kHz DC to 5 MHz (with MF-F5M option)
Output frequency resolution	1 μ Hz
Output phase range	$\pm 180^\circ$
Output phase resolution	10 μ deg
Differential outputs	Sine waves shifted by 180°
Output D/A conversion	16 bit, 60 MSa/s
Output amplitude ranges	± 10 mV, ± 100 mV, ± 1 V, ± 10 V (single ended into high impedance)
Output DC offset range	± 10 mV to ± 10 V, equal to the set output amplitude range
Output power	24 dBm (± 10 V, 250 mW), for each BNC
Output gain inaccuracy	< 1% at 100 kHz for all output ranges
Maximum output drive current	100 mA
Output offset amplitude	± 1 mV or 1% of range, whichever is bigger
Harmonic distortion	85 dBc for $f < 100$ kHz, 60 dBc for $f < 5$ MHz; for output ranges ≤ 1 V; 80 dBc for $f < 100$ kHz, 50 dBc for $f < 5$ MHz; for output range 10 V; carrier amplitude 1 dBFS
Analog adder	Auxiliary Input 1 can be added to the signal output, ± 10 V, DC–2 MHz

Table 5.12. Signal Output: voltage noise, ranges

Output range	Output noise density (high load impedance setting)	RMS output noise at 12 MHz bandwidth
10 mV	43 nV/ $\sqrt{\text{Hz}}$	145 μ Vrms
100 mV	43 nV/ $\sqrt{\text{Hz}}$	145 μ Vrms
1 V	48 nV/ $\sqrt{\text{Hz}}$	161 μ Vrms
10 V	104 nV/ $\sqrt{\text{Hz}}$	310 μ Vrms

Table 5.13. Auxiliary Inputs

Parameter	Description
Connectors	2 BNC on the front panel
A/D converter	16 bit, 15 MSa/s
A/D analog bandwidth	5 MHz
Input impedance	1 M Ω
Amplitude	± 10 V
Input noise amplitude	430 nV/ $\sqrt{\text{Hz}}$; frequency > 100 kHz
Resolution	0.335 mV

Table 5.14. Auxiliary Outputs

Parameter	Description
Connectors	4 BNC on the front panel
D/A converter	18 bit, 612 kSa/s
D/A analog bandwidth	200 kHz
Output impedance	50 Ω
Amplitude	± 10 V
Resolution	< 85 μ V
Drive current	20 mA
Noise density	210 nV/ $\sqrt{\text{Hz}}$ into high-impedance load; frequency > 1 kHz
RMS noise	90 μ V _{rms} into high-impedance load; measurement bandwidth 12 MHz

Table 5.15. Trigger Inputs

Parameter	Description
Connectors	2 BNC on the back panel
Trigger input impedance	1 k Ω
Frequency range external reference	1 Hz to 500 kHz; 1 Hz to 5 MHz, requires MF-F5M option
Trigger amplitude range	± 5 V
Minimum pulse width	35 ns
Trigger level	± 5 V, 3.66 mV resolution
Trigger hysteresis	< 20 mV

Table 5.16. Trigger Outputs

Parameter	Description
Connectors	2 BNC on the back panel
Trigger output impedance	50 Ω
Frequency range external reference	1 μ Hz to 500 kHz; 1 μ Hz to 5 MHz, requires MF-F5M option
Trigger amplitude	5 V

Table 5.17. 10 MHz clock synchronization

Parameter	Description
Connectors	2 BNC, 10 MHz clock input and output on the back panel
10 MHz input, impedance	50 Ω
10 MHz input, frequency range	9.98 to 10.02 MHz
10 MHz input, amplitude range	200 mV to 3 V
10 MHz output, impedance	50 Ω
10 MHz output, amplitude	Typically 1 V _{pp} into 50 Ω , sinusoidal

Table 5.18. Internal frequency reference

Parameter	Description
Type	TCXO
Initial accuracy	$< \pm 1.5$ ppm
Long term accuracy/aging	$< \pm 1$ ppm in the first year
Short term stability (0.1 s)	$< 2 \cdot 10^{-10}$
Temperature coefficient	0.05 ppm/°C (@23°C)
Phase noise at 1 kHz	-140 dBc/Hz
Phase noise at 10 kHz	-150 dBc/Hz

Table 5.19. Connectivity and others

Parameter	Description
Host connection	LAN, 1 GbE; USB 2.0, 480 Mbit/s
Internal drive data storage capacity	4.5 GB
USB host	2 connectors on the back panel for mass storage or WLAN modules
DIO, digital I/O	4 x 8 bit, general purpose digital input/output port, 3.3 V TTL VHDCI 68 pin female connector

Table 5.20. Maximum ratings

Parameter	Lower	Upper
Damage threshold Current Signal Input I	-5 V	+5 V
Damage threshold Voltage Input +V/-V Diff	-12 V	+12 V
Damage threshold Signal Output +V/-V	-12 V	+12 V
Damage threshold Aux Input 1,2	-12 V	+12 V
Damage threshold Aux Outputs 1,2,3,4	-12 V	+12 V
Damage threshold Clock 10 MHz In/Out	-5 V	+5 V
Damage threshold Trigger Out 1,2	-1 V	+6 V
Damage threshold Trigger In 1,2	-8 V	+8 V
Damage threshold DIO 32 bit	-1 V	+6 V
Damage threshold DC In	0 V	26 V

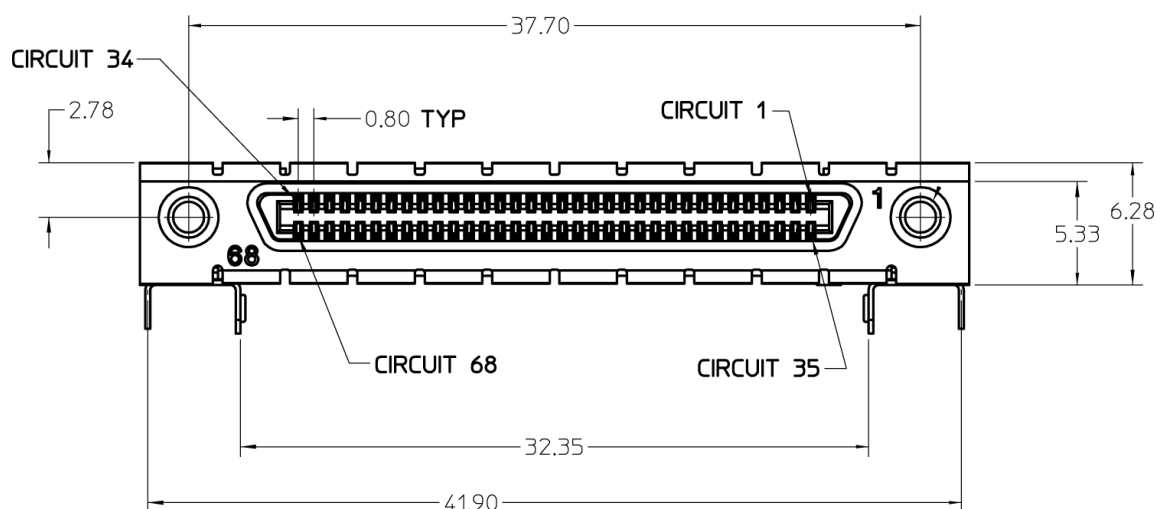
Table 5.21. LabOne UI requirements

Parameter	Description
Operating systems	Any, web browser based
Input device	Touch screen, keyboard, mouse
CPU	2+ cores, hardware accelerated rendering on browser
Browser	Internet Explorer 10+, Firefox 27+, Chrome 36+, Safari 6+, Opera 23+
Connectivity	1 GbE, 100 MbE, USB 2.0

Table 5.22. LabOne API requirements

Parameter	Description
Operating systems	Windows 10 (x64), Windows 8.1 (x64), Windows 7 (i386, x64), Linux Ubuntu 14.04 LTS (AMD64), 16.04 LTS (AMD64), Mac OS X
CPU	AMD K8 (Athlon 64, Sempron 64, Turion 64, etc.), AMD Phenom, Intel Pentium 4, Xeon, Celeron, Celeron D, Pentium M, Celeron M, Core, Core 2, Core i5, Core i7, Atom
RAM	4 GB+
Connectivity	1 GbE, 100 MbE, USB 2.0
Supported languages	LabVIEW 2009 (i386, x64) and later; Python 2.7, 3.5 (i386, AMD64); MATLAB 2009 (Windows) / MATLAB 2014 (Linux) and later; C/C++

The DIO port is a VHDCI 68 pin connector as introduced by the SPI-3 document of the SCSI-3 specification. It is a female connector that requires a 32 mm wide male connector. The DIO port features 32 bits that can be configured byte-wise as inputs or outputs.

**Figure 5.1. DIO HD 68 pin connector****Table 5.23. DIO pin assignment**

Pin	Name	Description	Range specification
68	CLKI	clock input, used to latch signals at the digital input ports - can also be used to retrieve digital signals from the output port using an external sampling clock	3.3 V LVCMOS/TTL
67	DOL	DIO output latch, 60 MHz clock signal, the digital outputs are synchronized to the falling edge of this signal	3.3 V LVCMOS/TTL
66-59	DI[31:24]	digital input or output (set by user)	3.3 V LVCMOS/TTL
58-51	DIO[23:16]	digital input or output (set by user)	3.3 V LVCMOS/TTL

Pin	Name	Description	Range specification
50-43	DIO[15:8]	digital input or output (set by user)	3.3 V LVCMOS/TTL
42-35	DIO[7:0]	digital input or output (set by user)	3.3 V LVCMOS/TTL
34-30	-	do not connect, for internal use only	-
29-1	GND	digital ground	-

The figure below shows the architecture of the DIO input/output. The DIO port features 32 bits that can be configured byte-wise as inputs or outputs by means of a drive signal. The digital output data is latched synchronously with the falling edge of the internal clock, which is running at 60 MHz. The internal sampling clock is available at the DOL pin of the DIO connector. Digital input data can either be sampled by the internal clock or by an external clock provided through the CLKI pin. A decimated version of the input clock is used to sample the input data. The Decimation unit counts the clocks to decimation and then latches the input data. The default decimation is 3e6, corresponding to a digital input sampling rate of 20 samples per second.

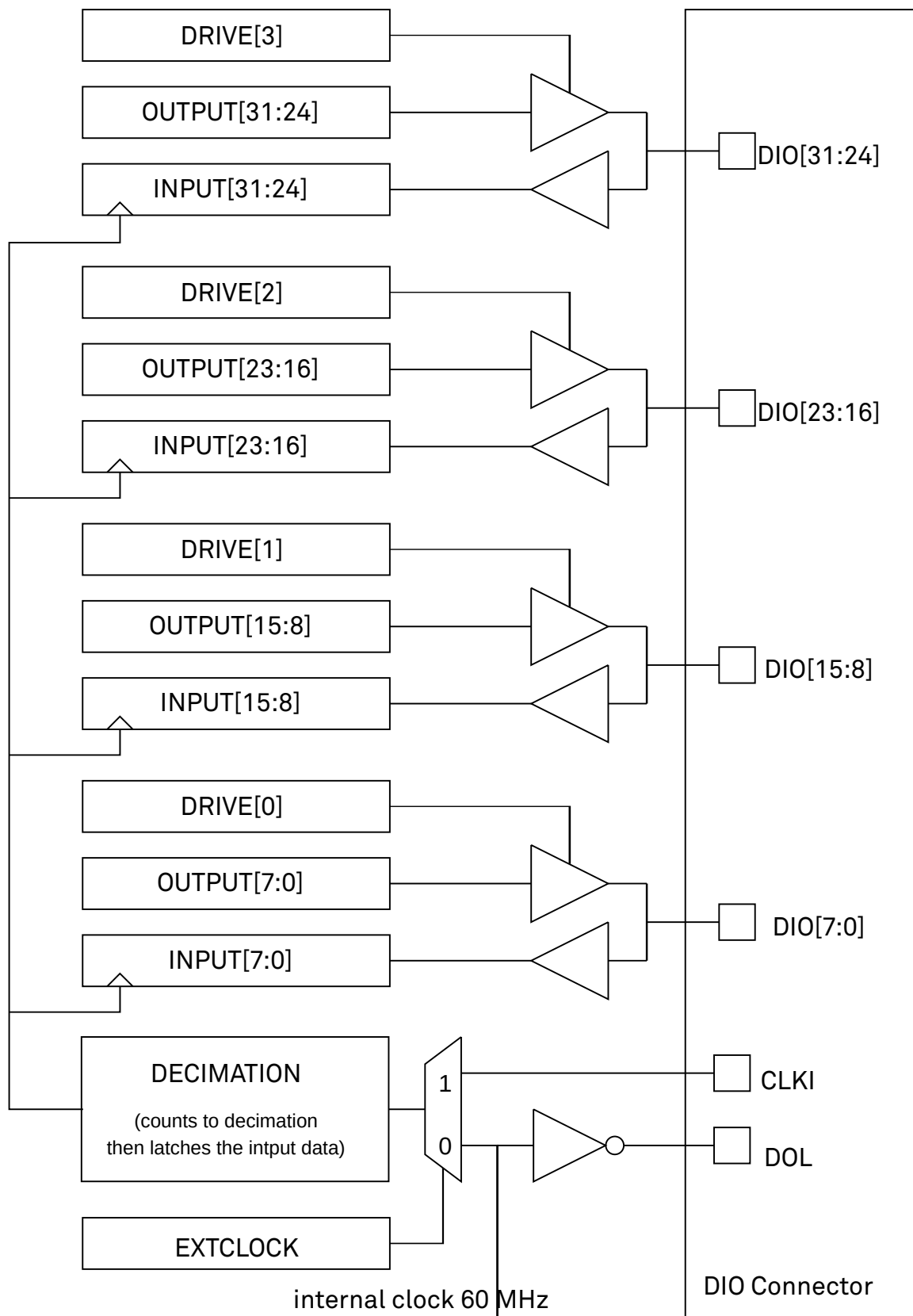


Figure 5.2. DIO input/output architecture

5.2. Performance Diagrams

Input noise amplitude depends on several parameters, and in particular on the frequency and on the input range setting. The input noise is lower for smaller input ranges, and it is recommended to use small ranges especially for noise measurements. [Figure 5.3](#) shows the noise density of the voltage input with DC coupling. The input noise with AC coupling is the same as long as the frequency is higher than the AC cutoff frequency (see [Table 5.8](#)). The noise is independent of the input impedance setting, 50 Ω or 10 M Ω . The corner frequency of the 1/f noise is in the range of 100 Hz and the white-noise floor is typically 2.5 nV/ $\sqrt{\text{Hz}}$ for the smallest input ranges.

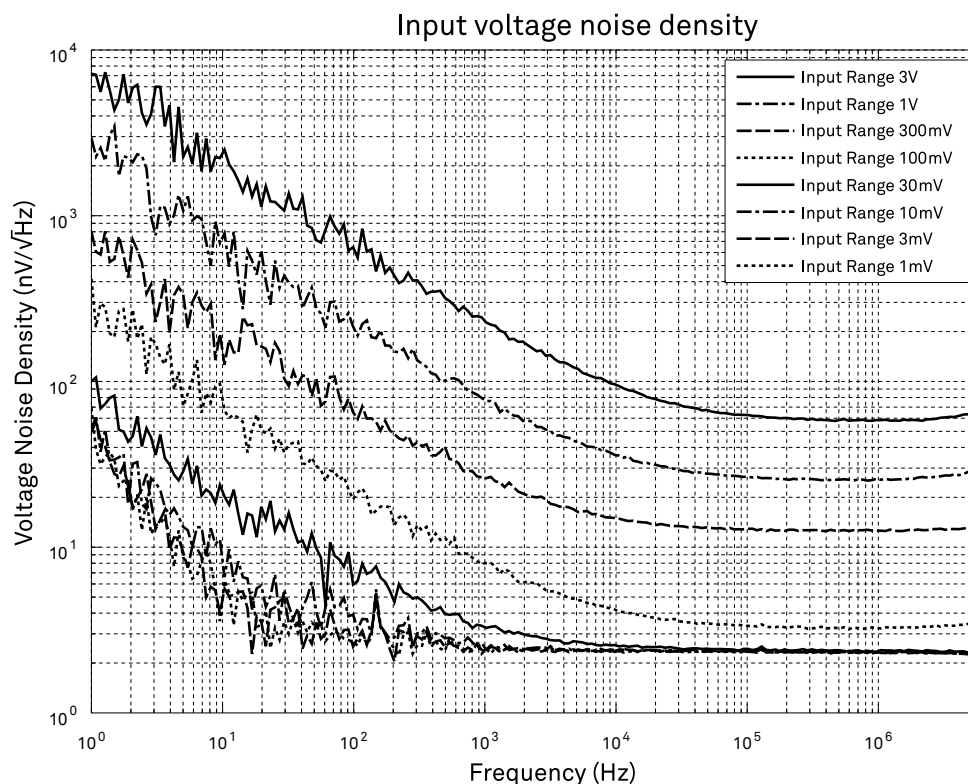


Figure 5.3. MFLI Voltage Input voltage noise density

A note on input voltage noise and AC coupling. The input voltage noise will increase at low frequencies, when using AC coupling at the signal input of the MFLI. For example, the input voltage noise of the 10 mV input range is around 10 nV/ $\sqrt{\text{Hz}}$ at a frequency of 10 Hz, as shown in the figure above. When switching on AC coupling, the noise increases to around 75 nV/ $\sqrt{\text{Hz}}$. The reason is that the 10nF AC coupling capacitor has an impedance of 1.6 M Ω at 10 Hz. Together with the 10 M Ω input bias resistor of the MFLI input amplifier, you get the increased noise density. To work around this problem, use DC coupling for the signal input. An external large capacitor may be employed if AC coupling must be used.

[Figure 5.4](#) shows the noise density of the current input. The frequency range of the measurements for the smaller input ranges is limited by the bandwidth of these ranges as specified in [Table 5.10](#).

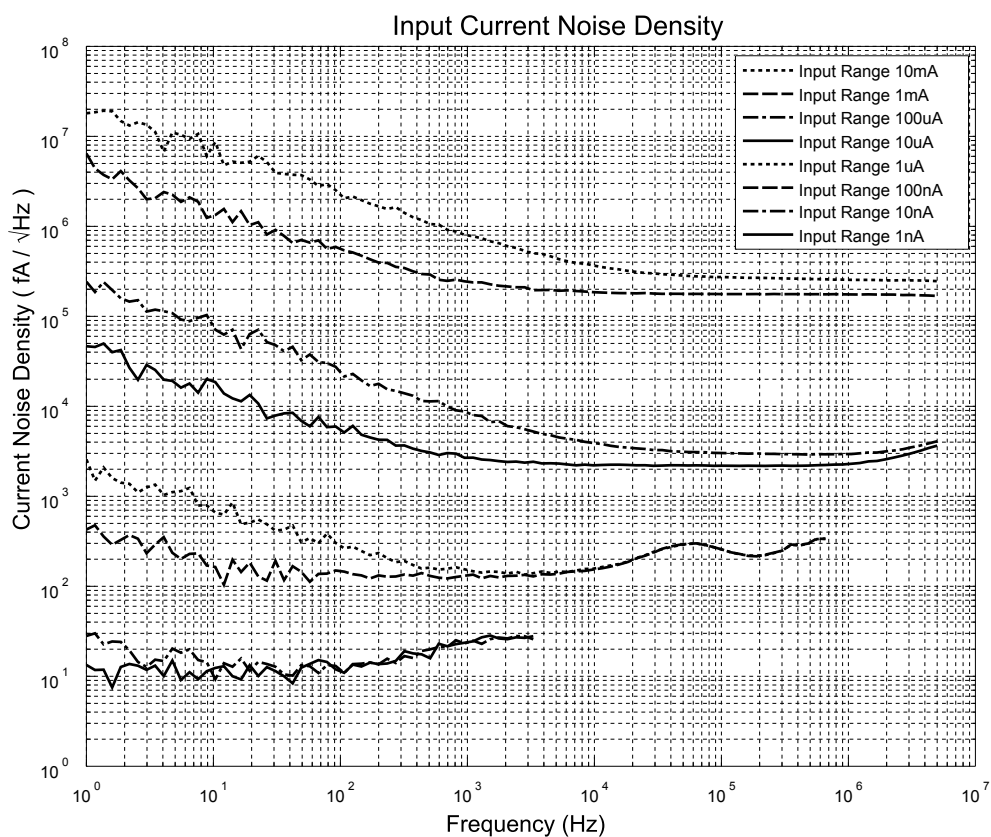


Figure 5.4. MFLI Current Input current noise density

Figure 5.5 shows the SSB phase noise measured at the signal output. For this measurement, the signal output was connected to a phase noise analyzer and the output amplitude was set to 5 V. The measured phase noise at 5 MHz and 10 kHz offset is around -153 dBc/Hz.

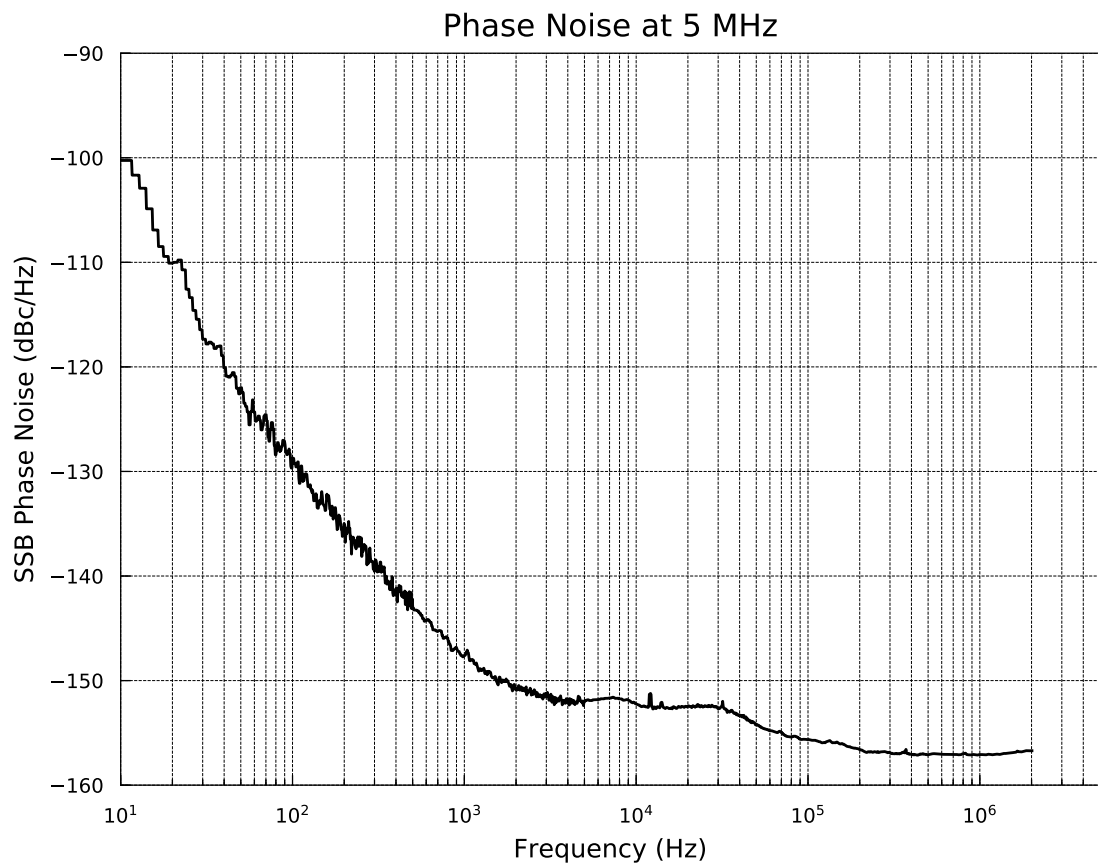


Figure 5.5. MFLI phase noise

Figure 5.6 shows the noise density of the voltage output into 50 Ω . For high-impedance loads, the voltage noise levels in the figure need to be doubled. The output noise was measured with no signal and 0 V offset. For the ranges 10 mV and 100 mV, the voltage noise is basically identical with a flat-band noise level of 14.5 nV/ $\sqrt{\text{Hz}}$ at 100 kHz into 50 Ω . For high-impedance loads, the value would be around 29 nV/ $\sqrt{\text{Hz}}$. The 10 V range shows the largest noise with 34.5 nV/ $\sqrt{\text{Hz}}$ at 100 kHz. The corner frequency of the 1/f noise is in the range of 10 to 20 kHz.

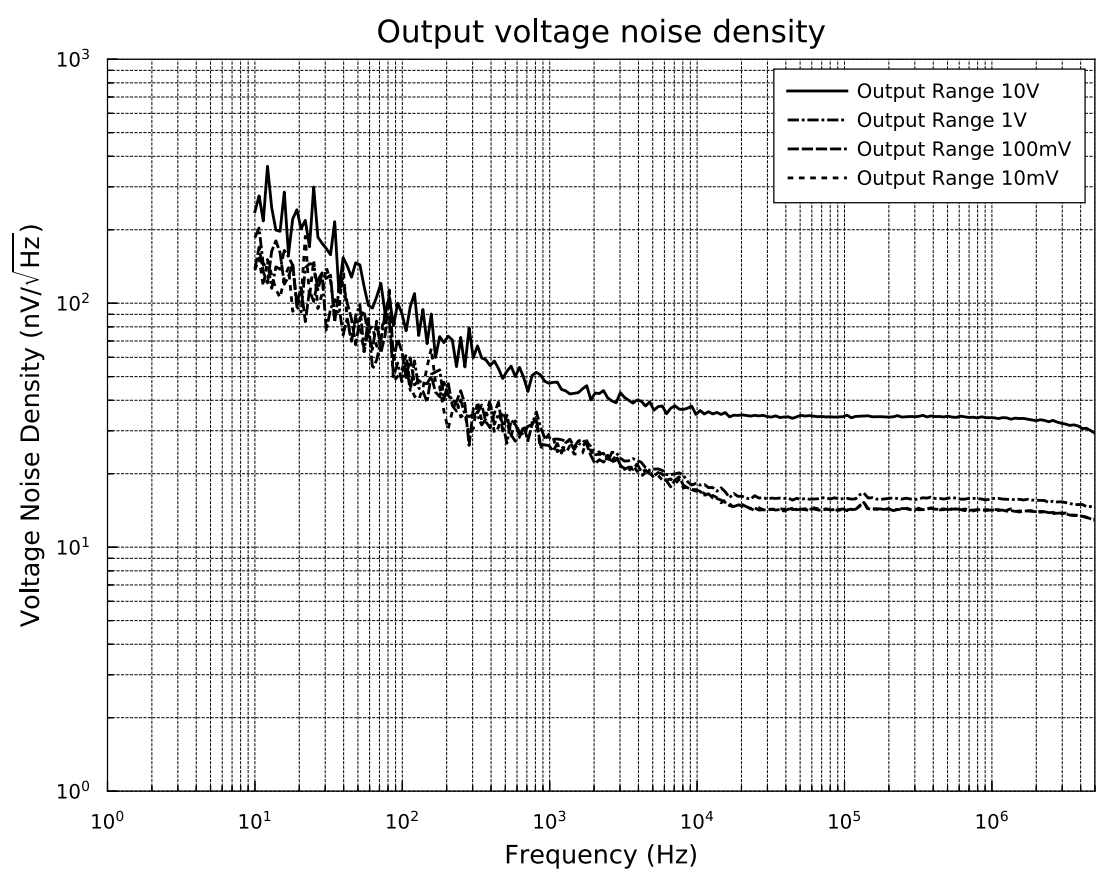


Figure 5.6. MFLI Signal Output voltage noise density